

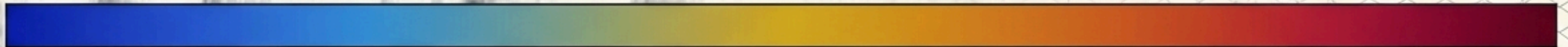
Mapping Acoustic Signal Features to Neural Activation Patterns During Music Listening Using fMRI

Biomedical Image Analysis - Project

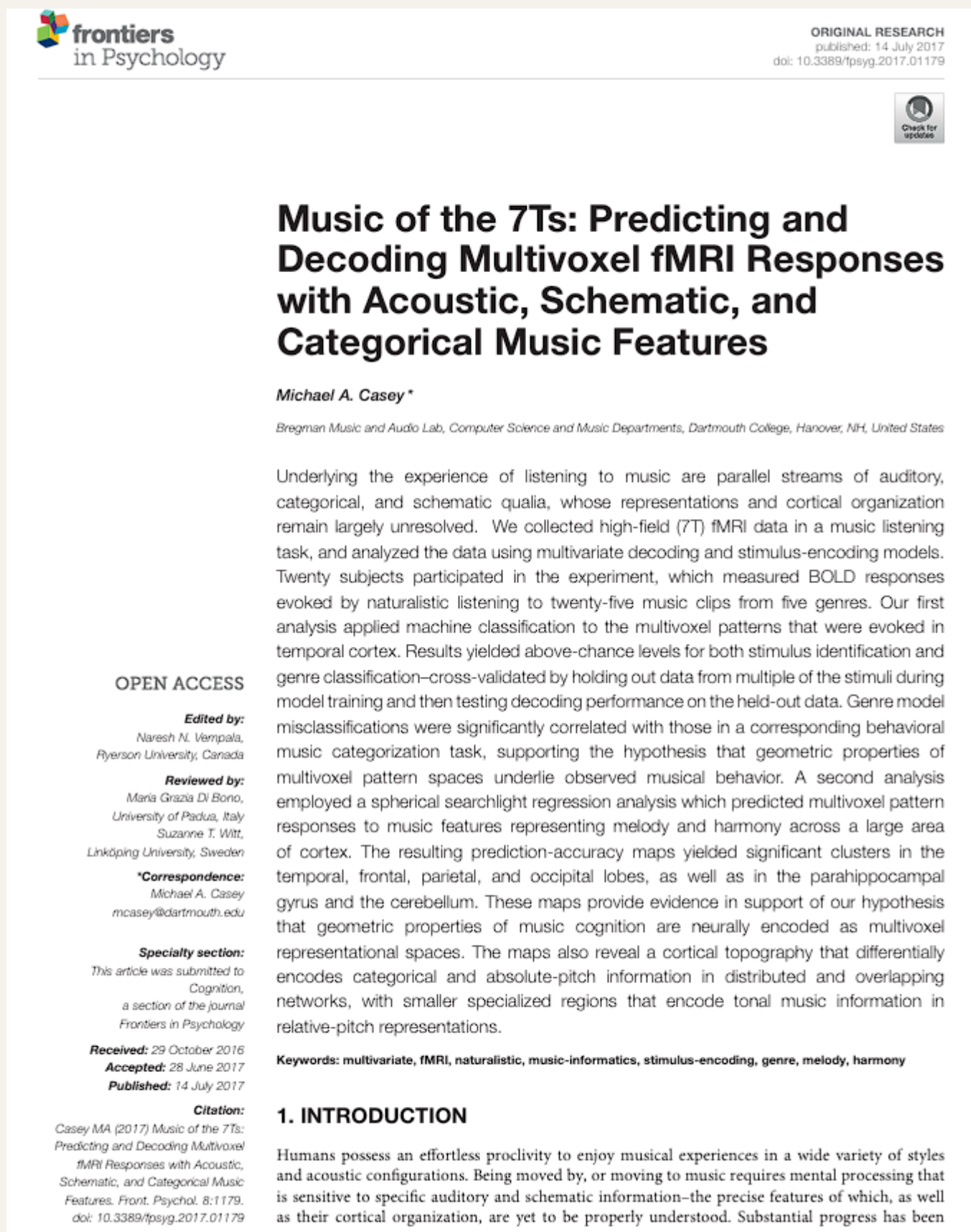
Shreepal - EC23B1107

Shriya - EC23I1008

Ashok - EC23I1014



Journal Review



<https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2017.01179/full>

Dataset:- <https://openfmri.org/dataset/ds000113b/>

Dataset Details -

- Participants listened to clips from multiple music genres
- Whole-brain BOLD fMRI recordings captured during listening.

Data includes -

- BIDS(Brain Imaging Data Structure) standard dataset
- fMRI scans
- Stimulus timing information
- Audio files used in the experiment
- Preprocessing metadata

Journal Review

F1000Research

F1000Research 2015, 4:174 Last updated: 24 DEC 2025

Check for updates

DATA NOTE

High-resolution 7-Tesla fMRI data on the perception of musical genres – an extension to the *studyforrest* dataset

[version 1; peer review: 2 approved with reservations]

Michael Hanke ^{1,2}, Richard Dinga¹, Christian Häusler ¹, J. Swaroop Guntupalli³, Michael Casey⁴, Falko R. Kaule^{1,5}, Jörg Stadler⁶

¹Psychoinformatics lab, Department of Psychology II, University of Magdeburg, Magdeburg, Germany

²Center for Behavioral Brain Sciences, Magdeburg, Germany

³Department of Psychological and Brain Sciences, Dartmouth College, Hanover, New Hampshire, USA

⁴Bregman Laboratory, Department of Music and Department of Computer Science, Dartmouth College, Hanover, New Hampshire, USA

⁵Visual Processing Laboratory, Ophthalmic Department, Otto-von-Guericke-University, Magdeburg, Germany

⁶Leibniz Institute for Neurobiology, Magdeburg, Germany

v1

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Abstract

Here we present an extension to the *studyforrest* dataset – a versatile resource for studying the behavior of the human brain in situations of real-life complexity (<http://studyforrest.org>). This release adds more high-resolution, ultra high-field (7 Tesla) functional magnetic resonance imaging (fMRI) data from the same individuals. The twenty participants were repeatedly stimulated with a total of 25 music clips, with and without speech content, from five different genres using a slow event-related paradigm. The data release includes raw fMRI data, as well as precomputed structural alignments for within-subject and group analysis. In addition to fMRI, simultaneously recorded cardiac and respiratory traces, as well the complete implementation of the stimulation paradigm, including stimuli, are provided. An initial quality control analysis reveals distinguishable patterns of response to individual genres throughout a large expanse of areas known to be involved in auditory and speech processing. The present data can be used to, for example, generate encoding models for music perception that can be validated against the previously released fMRI data from stimulation with the “Forrest Gump” audio-movie and its rich musical content. In order to facilitate replicative and derived works, only free and open-source software was utilized.

Open Peer Review

Approval Status  

	1	2
version 1 29 Jun 2015	 view	 view

1. Cristiano Micheli, School of Medicine and Health Sciences, University of Oldenburg, Oldenburg, Germany

Jochem Rieger, University of Oldenburg, Oldenburg, Germany

Kirsten Weber, Max Planck Institute for Psycholinguistics, Nijmegen, The Netherlands

2. Karsten Müller, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany

Any reports and responses or comments on the article can be found at the end of the article.

<https://f1000research.com/articles/4-174/v1>

Journal Review

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Article | [Open access](#) | Published: 02 February 2018

Identifying musical pieces from fMRI data using encoding and decoding models

[Sebastian Hoefle](#), [Annerose Engel](#), [Rodrigo Basilio](#), [Vino Alluri](#), [Petri Toivainen](#), [Maurício Cagy](#) & [Jorge](#)

[Moll](#) 

[Scientific Reports](#) **8**, Article number: 2266 (2018) | [Cite this article](#)

12k Accesses | 43 Citations | 238 Altmetric | [Metrics](#)

Abstract

Encoding models can reveal and decode neural representations in the visual and semantic domains. However, a thorough understanding of how distributed information in auditory cortices and temporal evolution of music contribute to model performance is still lacking in the musical domain. We measured fMRI responses during naturalistic music listening and constructed a two-stage approach that first mapped musical features in auditory cortices and then decoded novel musical pieces. We then probed the influence of stimuli duration (number of time points) and spatial extent (number of voxels) on decoding accuracy. Our approach revealed a linear increase in accuracy with duration and a point of optimal model performance for the spatial extent. We further showed that Shannon entropy is a driving factor, boosting accuracy up to 95% for music with highest information content. These findings provide key insights for future decoding and reconstruction algorithms and open new venues for possible clinical applications.

ORIGINAL RESEARCH

Brain and Behavior  WILEY

Correspondence of categorical and feature-based representations of music in the human brain

Tomoya Nakai^{1,2}  | Naoko Koide-Majima^{2,3}  | Shinji Nishimoto^{1,2,4} 

¹Center for Information and Neural Networks, National Institute of Information and Communications Technology, Suita, Japan

²Graduate School of Frontier Biosciences, Osaka University, Suita, Japan

³AI Science Research and Development Promotion Center, National Institute of Information and Communications Technology, Suita, Japan

⁴Graduate School of Medicine, Osaka University, Suita, Japan

Correspondence

Shinji Nishimoto, Center for Information and Neural Networks, National Institute of Information and Communications Technology, Suita, Osaka, Japan. Email: nishimoto@nict.go.jp

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Abstract

Introduction: Humans tend to categorize auditory stimuli into discrete classes, such as animal species, language, musical instrument, and music genre. Of these, music genre is a frequently used dimension of human music preference and is determined based on the categorization of complex auditory stimuli. Neuroimaging studies have reported that the superior temporal gyrus (STG) is involved in response to general music-related features. However, there is considerable uncertainty over how discrete music categories are represented in the brain and which acoustic features are more suited for explaining such representations.

Methods: We used a total of 540 music clips to examine comprehensive cortical representations and the functional organization of music genre categories. For this purpose, we applied a voxel-wise modeling approach to music-evoked brain activity measured using functional magnetic resonance imaging. In addition, we introduced a novel technique for feature-brain similarity analysis and assessed how discrete music categories are represented based on the cortical response pattern to acoustic features.

Results: Our findings indicated distinct cortical organizations for different music genres in the bilateral STG, and they revealed representational relationships between different music genres. On comparing different acoustic feature models, we found that these representations of music genres could be explained largely by a biologically plausible spectro-temporal modulation-transfer function model.

Conclusion: Our findings have elucidated the quantitative representation of music genres in the human cortex, indicating the possibility of modeling this categorization of complex auditory stimuli based on brain activity.

KEYWORDS

fMRI, MTF model, music genre, STG

Patient consent: Informed consent was obtained from all participants prior to their participation.

Permission to reproduce material from other sources: Not applicable.

Clinical trial registration: Not applicable.

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HUMAN BRAIN MAPPING

► *Hum Brain Mapp.* 2017 Mar 28;38(6):2955–2970. doi: [10.1002/hbm.23565](https://doi.org/10.1002/hbm.23565) 

Connectivity patterns during music listening: Evidence for action-based processing in musicians

[Vino Alluri](#)^{1,*} , [Petri Toivainen](#)^{1,*}, [Iballe Burunat](#)¹, [Marina Kliuchko](#)², [Peter Vuust](#)³, [Elvira Brattico](#)^{3,4}

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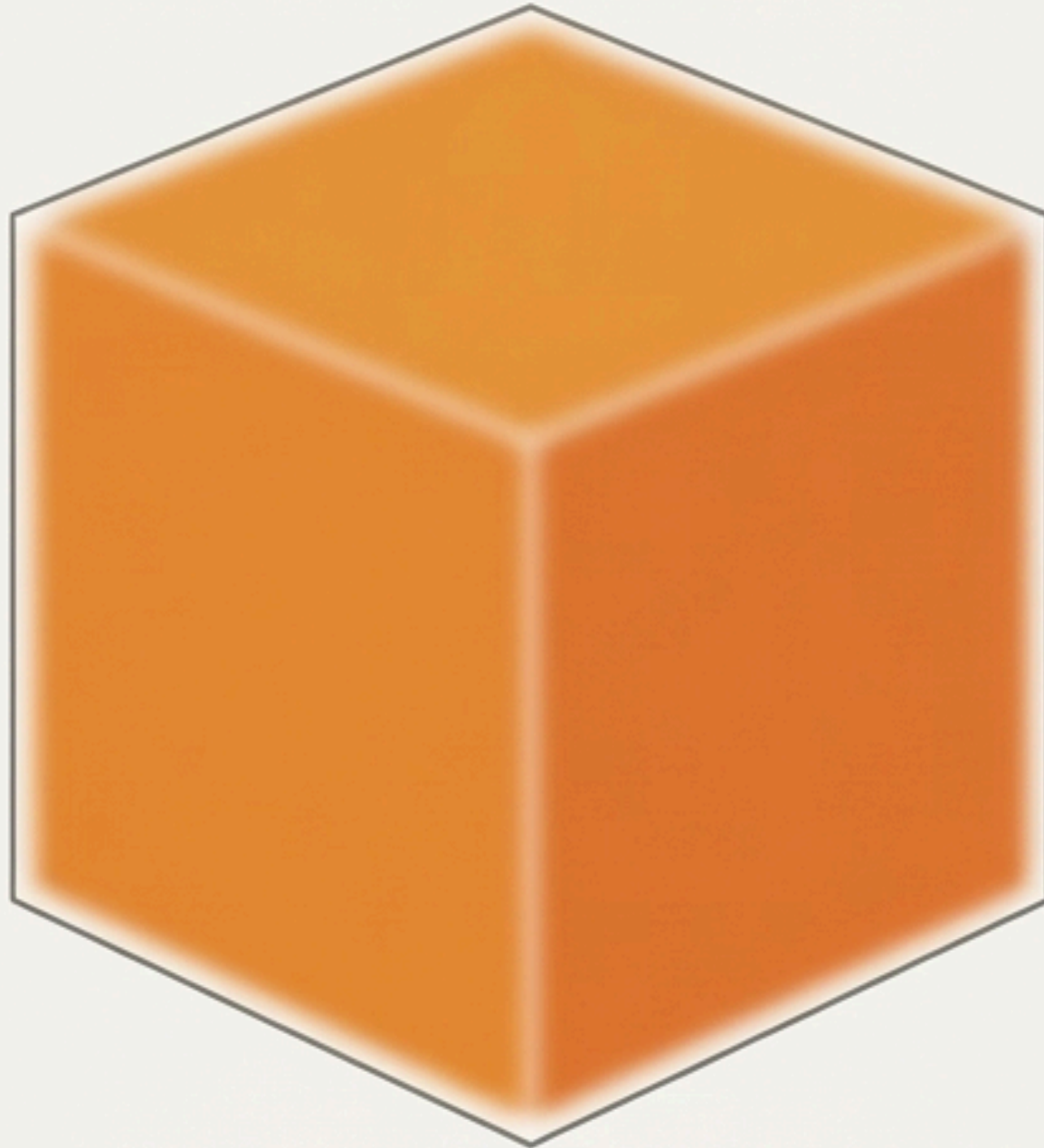
PMCID: PMC6866725 PMID: [28349620](#)

Abstract

Musical expertise is visible both in the morphology and functionality of the brain. Recent research indicates that functional integration between multi-sensory, somato-motor, default-mode (DMN), and salience (SN) networks of the brain differentiates musicians from non-musicians during resting state. Here, we aimed at determining whether brain networks

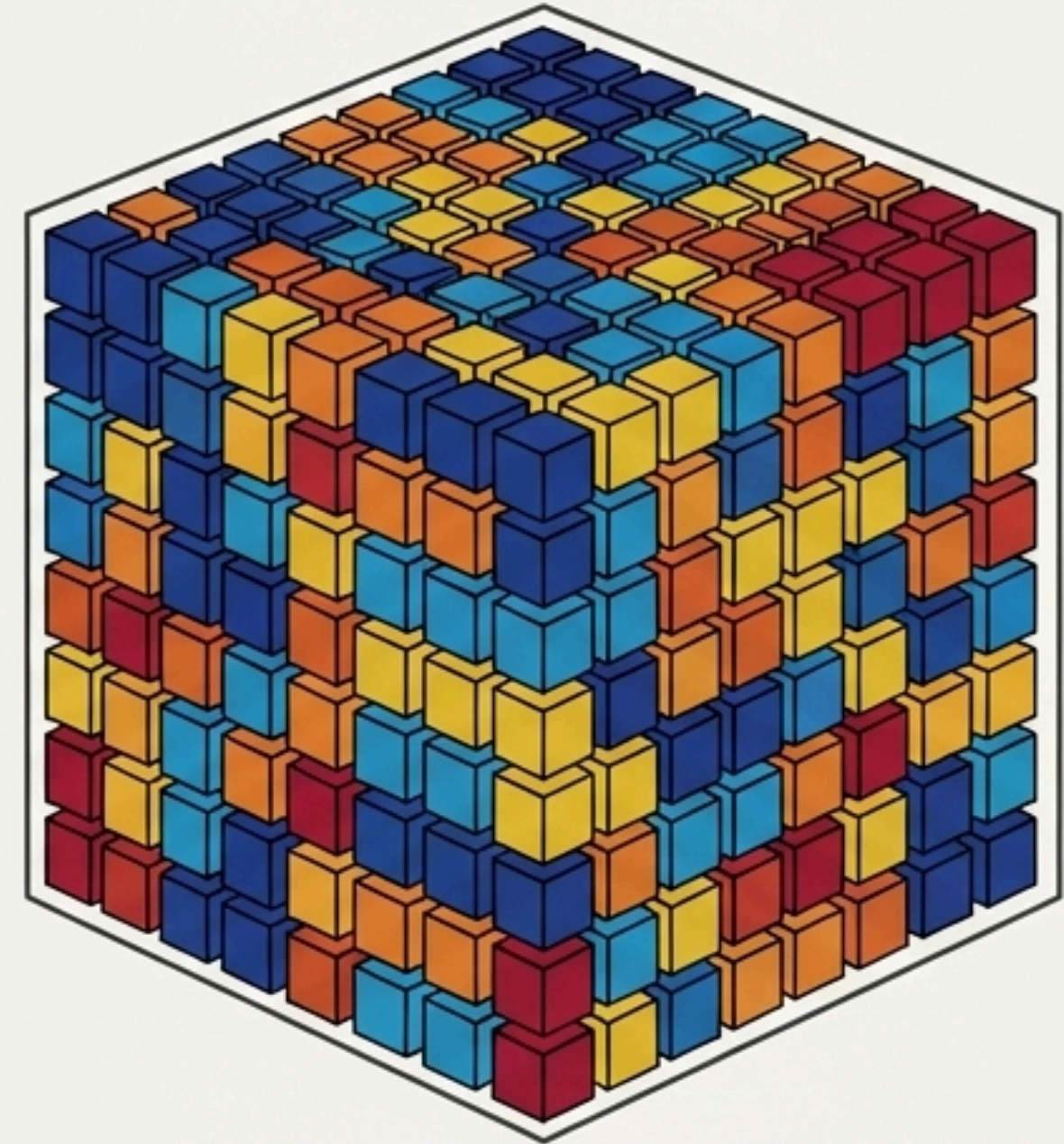
The 7-Tesla Advantage for Pattern Recognition

Standard Resolution: Gross Regional Activation



Traditional fMRI identifies general areas of activity but merges distinct neural population codes into single data points.

Ultra-High-Field 7T: Voxel-Level Pattern Topography



At 1.4 mm in-plane resolution, 7T reveals the highly detailed, overlapping multivoxel patterns required to mathematically separate complex stimuli like musical genre.

The Naturalistic Auditory Stimulus Matrix



Ambient

Seed: Brian Eno

Minimal rhythmic
transient markers.



Country

Seed: Waylon
Jennings



Heavy Metal

Seed: Ozzy
Osbourne



Rock'n'Roll

Seed: Eddie
Cochran



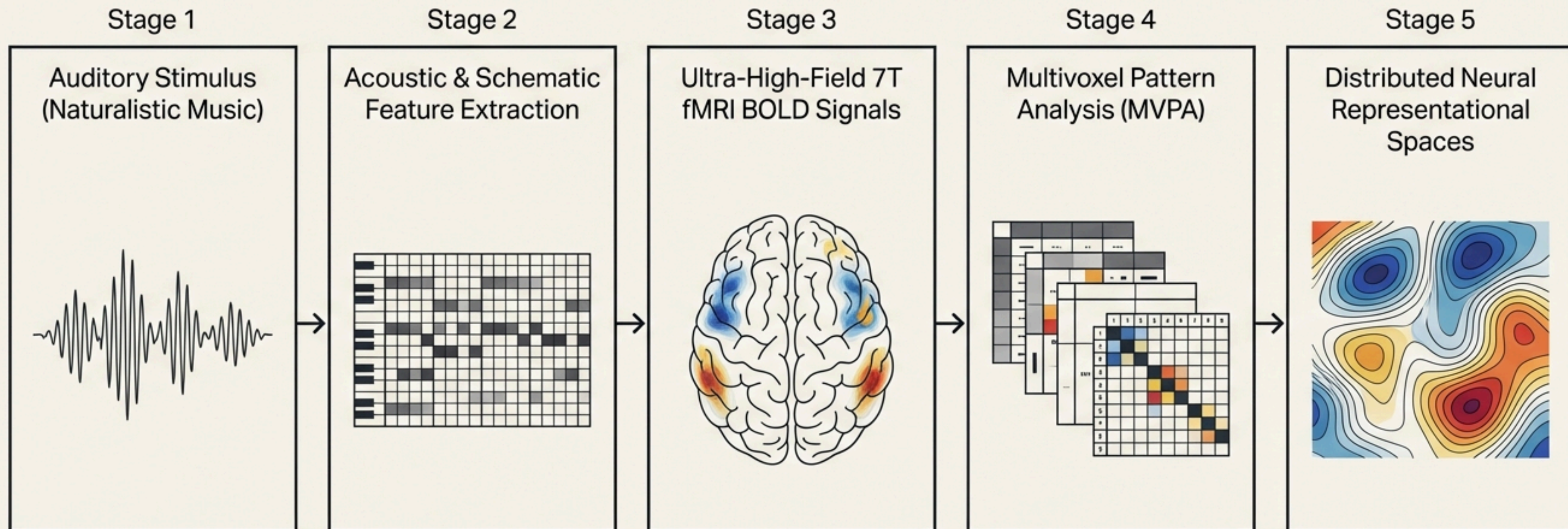
Symphonic

Seed: Beethoven

Purely
instrumental,
dense harmonic
structures.

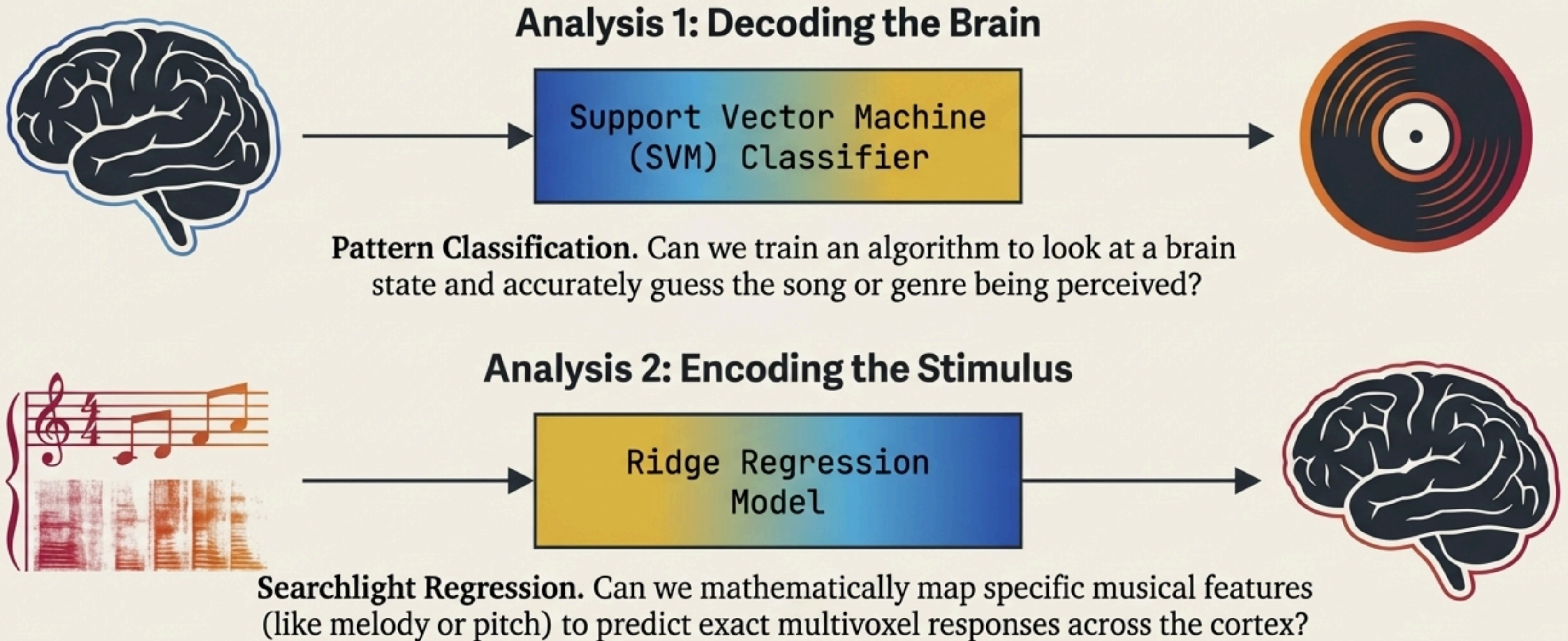
25 distinct 6-second clips | 44.1 kHz stereo | RMS power-balanced |
De Bruijn cycle sequence randomization to prevent temporal familiarity bias.

The Architecture of Musical Perception



Core Insight: Human musical experience relies on parallel streams of auditory, categorical, and schematic qualia. Understanding this requires $\neq$ multidimensional geometric spaces within the cortex.

“Bi-Directional Mapping of the Musical Mind” in GT America

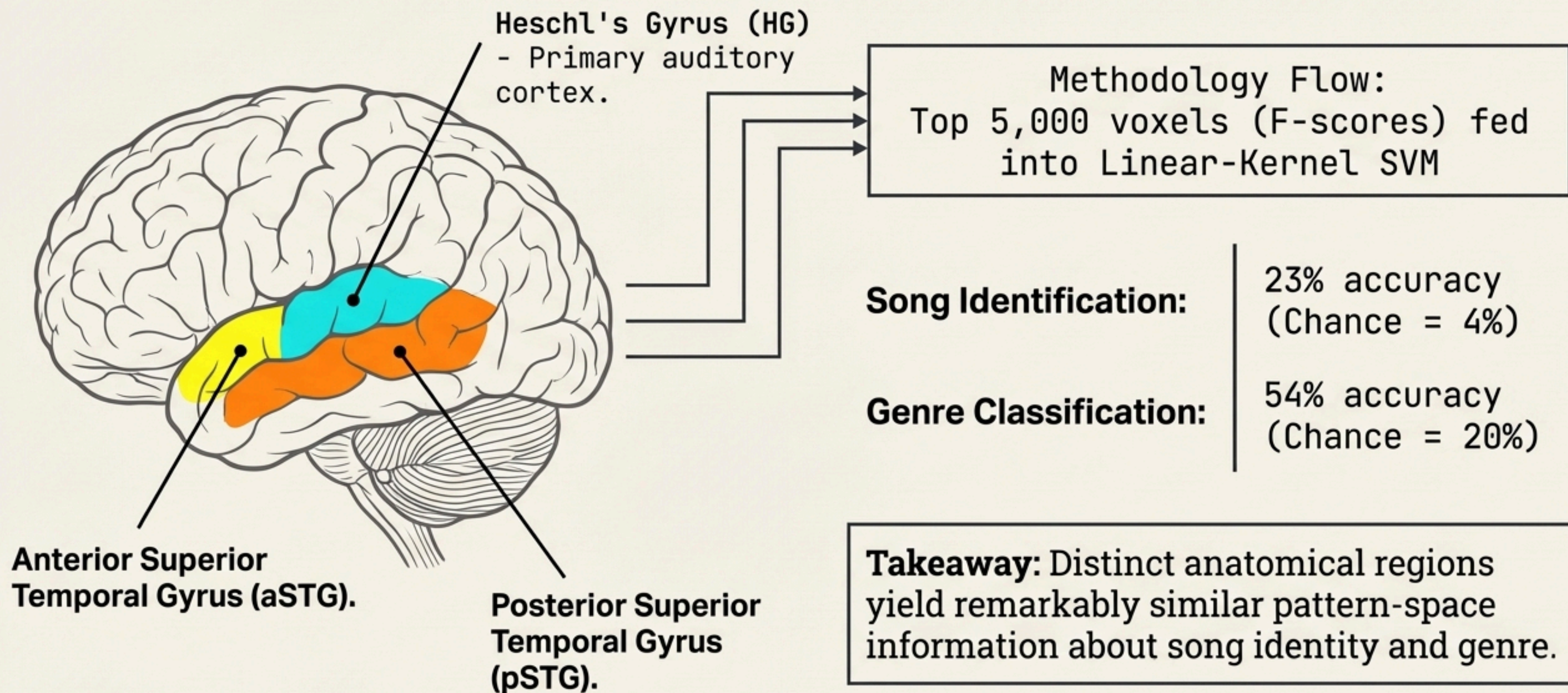


Synthesis: Decoding proves the brain categorizes music.
Encoding reveals how and where those exact features are distributed.

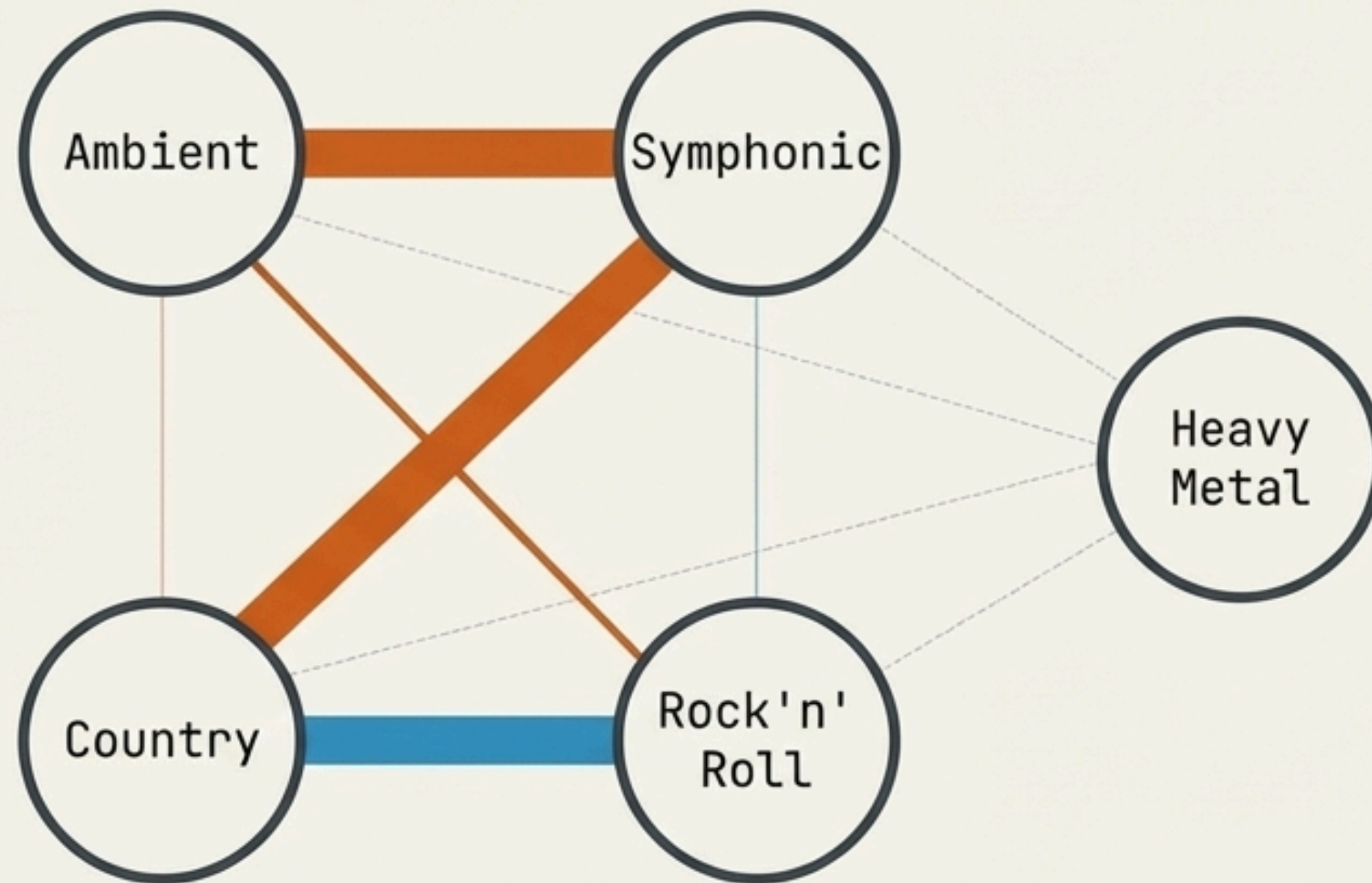
Pre-possesing

- removal of facial information from high-resolution anatomical image (code: `deface.py`)
- motion correction using FSL `mcflirt` tool (code: `run_mcflirt.py`)
- brain extraction for fMRI data using FSL `bet` tool (code: `run_betfunc.py`)
- brain extraction for inplane anatomical images using FSL `bet` tool (code: `bet_inplane.py`)
- convert high-resolution anatomy image into freesurfer format (code: `fs_setup.py`)
- brain extraction and cortical reconstruction for high-resolution anatomical images using freesurfer (code: `run_autorecon.py`)
- copy brain-extracted high-resolution anatomy back into main subject directory (code: `run_copy_stripped.py`)
- creation of first-level design files and analysis using FSL `feat` (code: `mk_level1_fsf.py`, `mk_all_level1_fsf.py`, `design_level1.stub`)
- Design details are identified from onset files in the model directories as well as key files in the main models directory
- Motion parameters are automatically included in the model as nuisance regressors

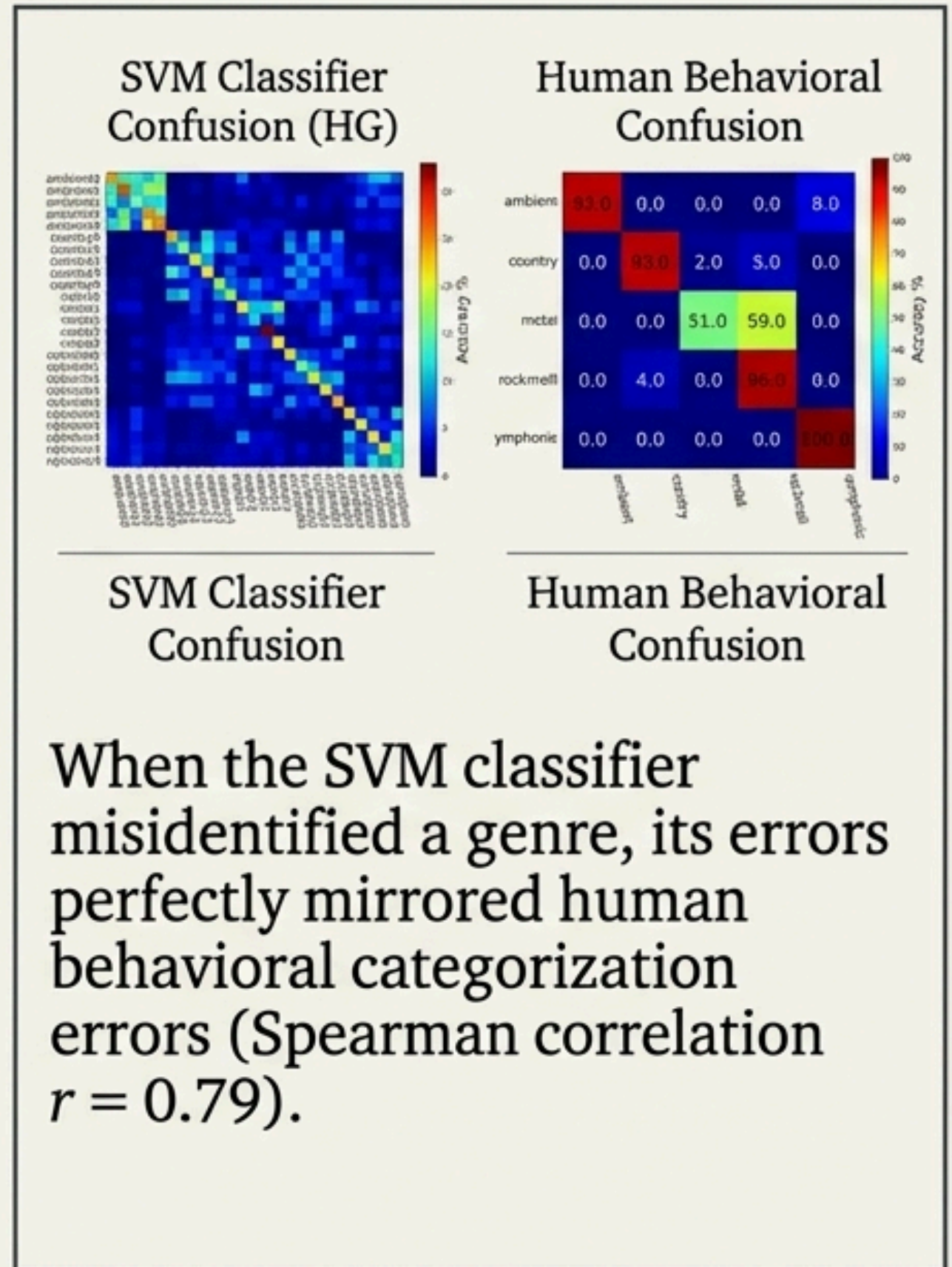
Analysis 1: Predicting Genre from Temporal Cortex Patterns



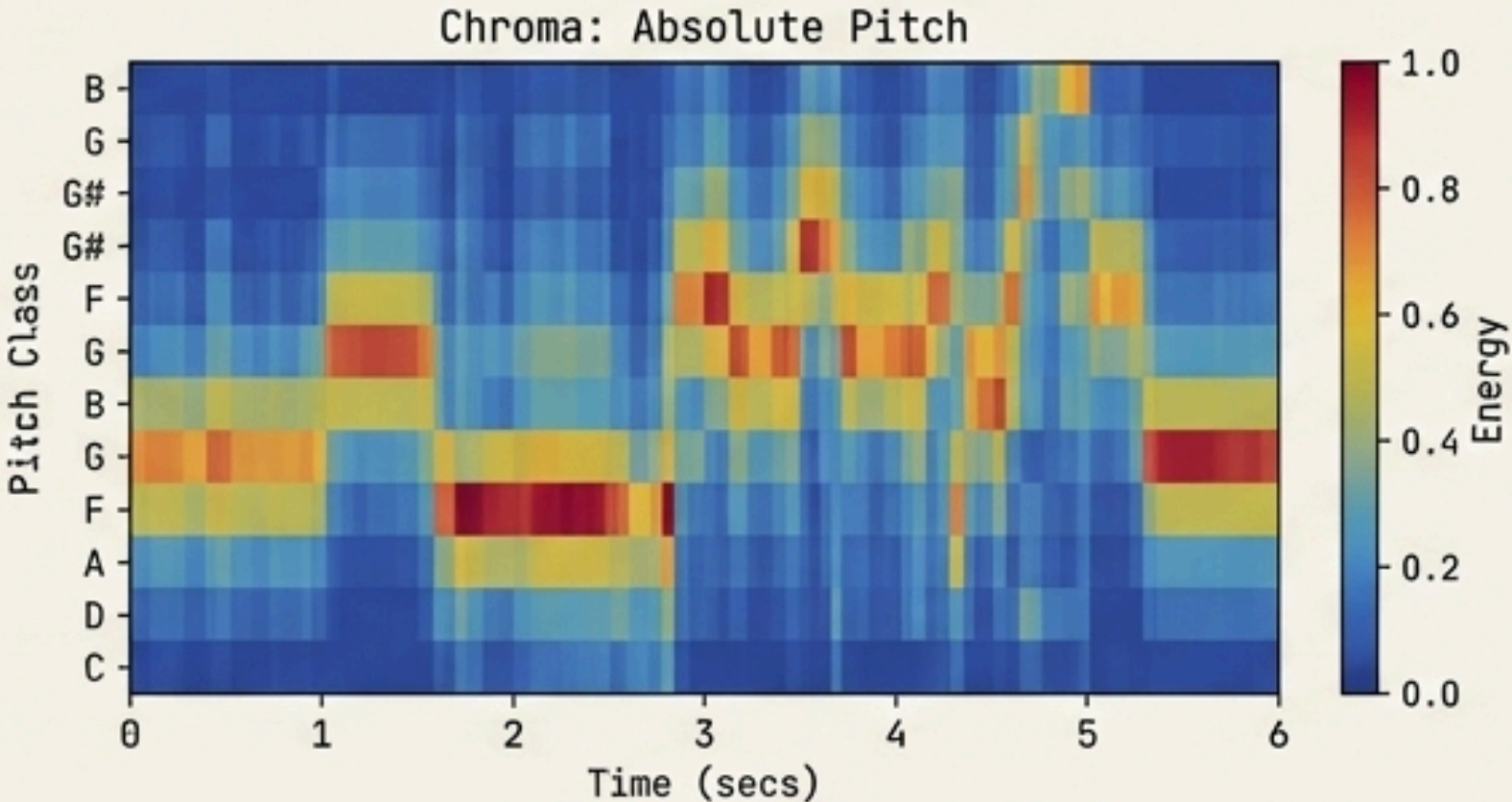
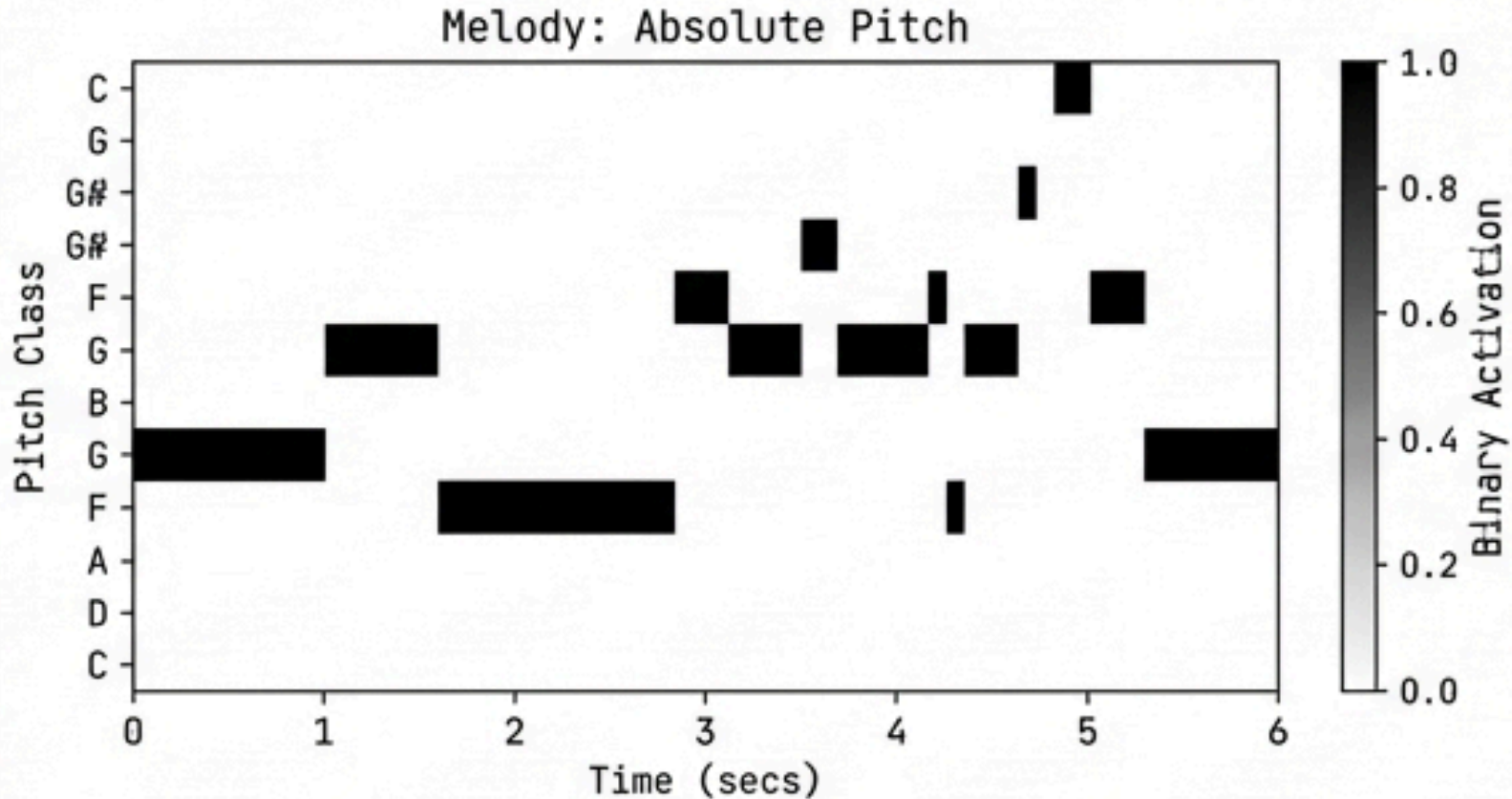
The Network of Confusion: Aligning Machine and Human Error



Synthesized Insight: The models and humans confused the same genres, implying the brain relies on super-ordinate categories—like the fundamental presence or absence of vocals and percussion—before distinguishing nuanced genre tags.



The Feature Space Matrix: Quantifying Music

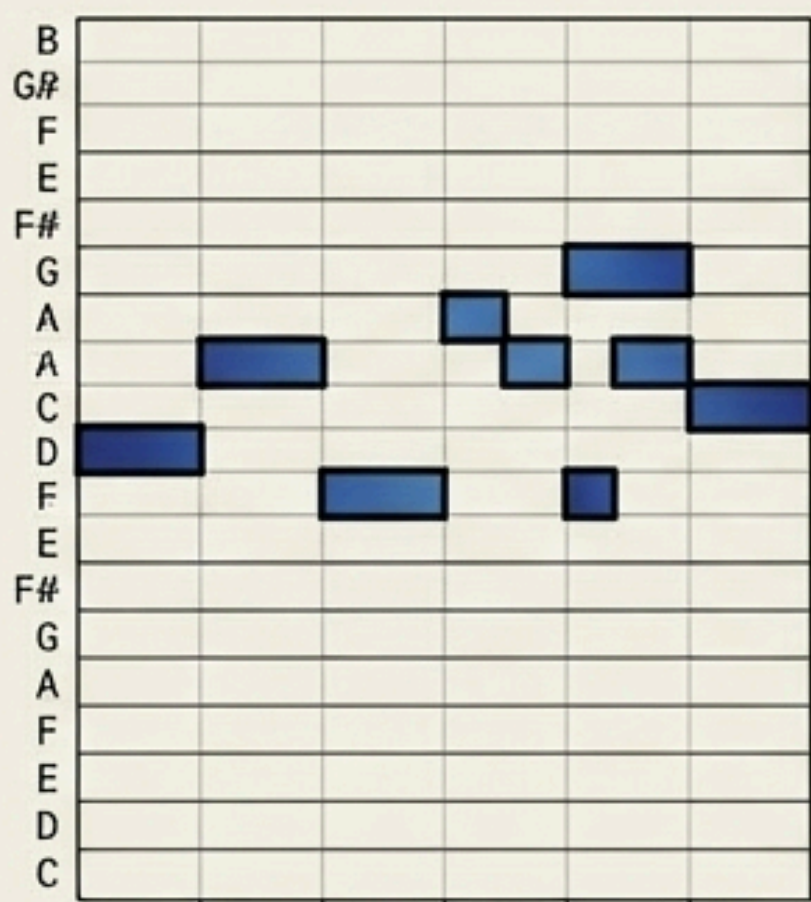
Category A: Acoustic Features (The Raw Sound)	Category B: Schematic Features (The Abstract Concept)
Examples: Mel-Frequency Spectrum (MFS), Cepstral Coefficients (MFCC), Audio Chromagrams.	Examples: Melody (Absolute Pitch), Melody (Relative Pitch).
Chroma: Absolute Pitch  The heatmap displays energy levels for 12 pitch classes (B, G, G#, G#, F, G, B, G, F, A, D, C) over a 6-second duration. The color scale ranges from 0.0 (blue) to 1.0 (red). A prominent horizontal band of high energy (yellow/red) is visible between 1 and 3 seconds, centered around the F and G pitch classes.	Melody: Absolute Pitch  The heatmap displays binary activation (0.0 to 1.0) for 12 pitch classes (C, G, G#, G#, F, G, B, G, F, A, D, C) over a 6-second duration. The color scale ranges from 0.0 (white) to 1.0 (black). The plot shows a sequence of discrete activations, with a notable pattern of high activation (black) between 1 and 3 seconds, centered around the G and F pitch classes.

Core Question: Does the brain encode the raw acoustic energy of a song, or the abstract mathematical relationship of its notes?

Isolating Musical Abstraction: Absolute vs. Relative Pitch

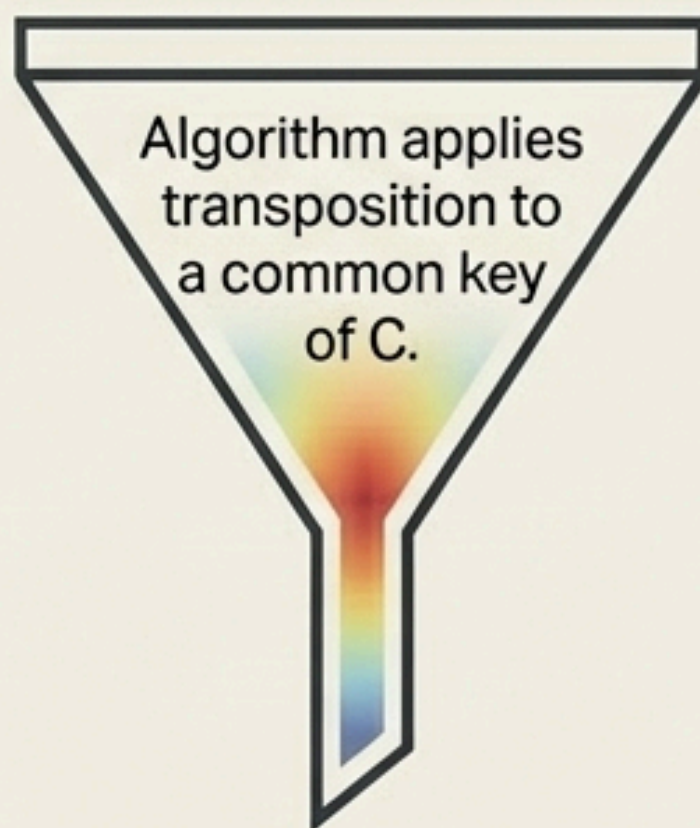
Transposition Engine

The Raw Input: Absolute Pitch

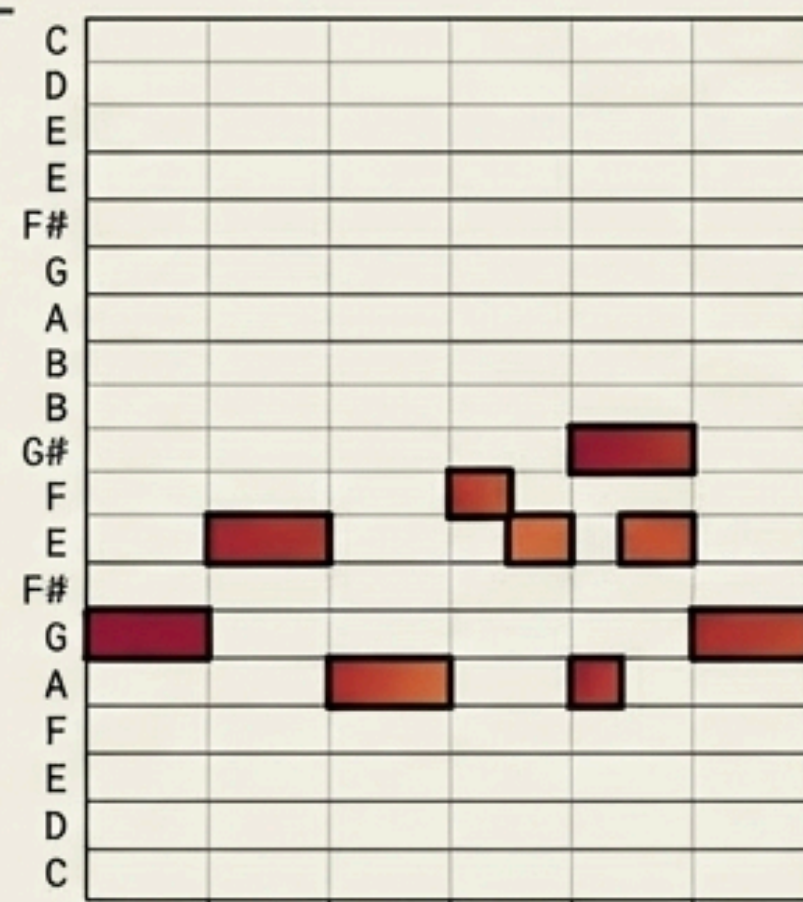


Tied strictly to specific physical frequencies (Hz) in the physical world.

The Transformation



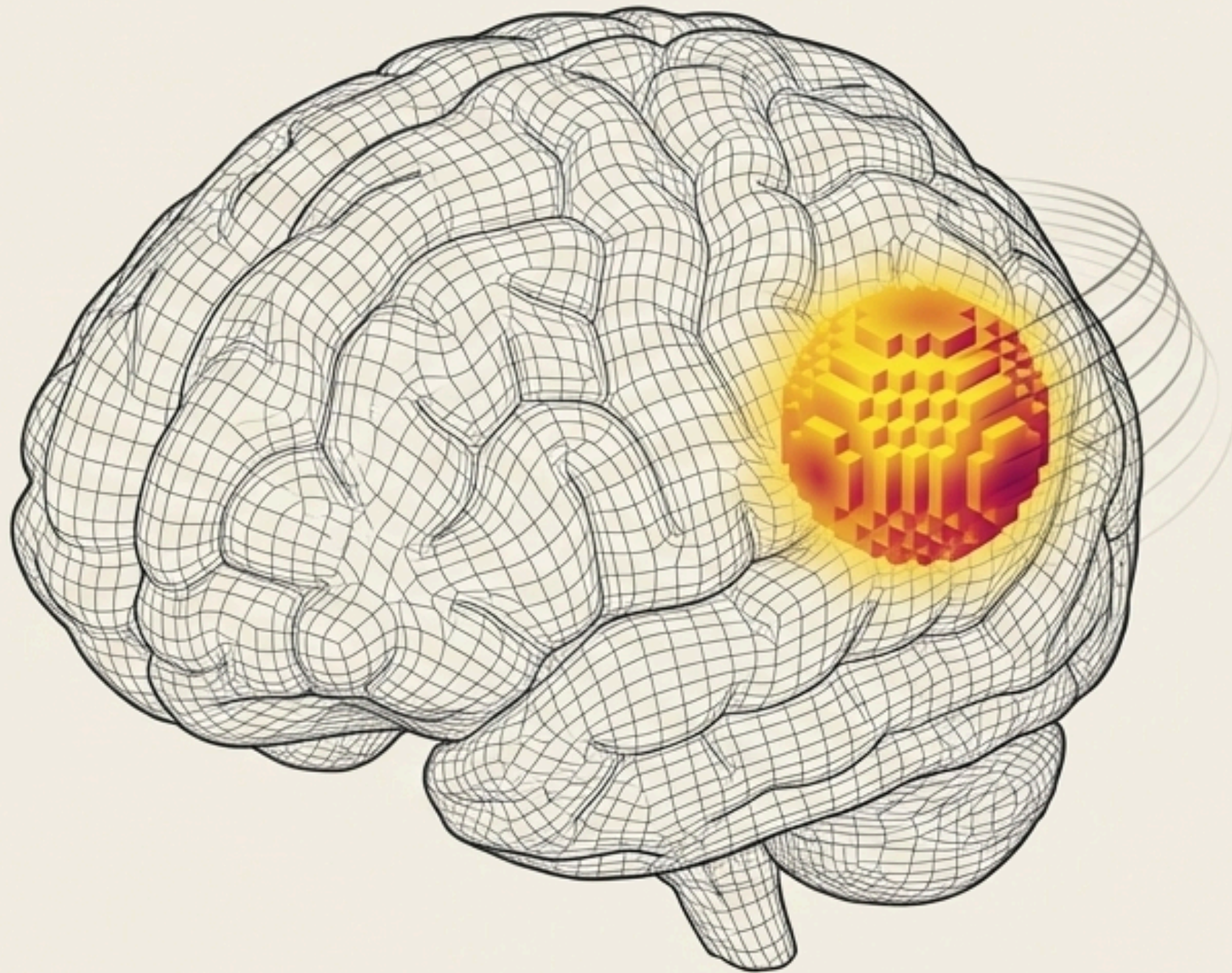
The Schema: Relative Pitch



Interval relationships are mapped independent of original frequency.

Connecting Insight: Relative pitch encoding requires the brain to process geometric interval relationships regardless of the actual frequencies hitting the eardrum. Finding this in the brain proves a higher level of cognitive musical abstraction.

Analysis 2: The Spherical Searchlight Algorithm



1. Isolate

Group a tiny 3-voxel sphere of brain tissue (max 123 voxels).



3. Score & Shift

Apply Tikhonov-regularized Ridge Regression to predict the voxel pattern based strictly on the extracted musical features (Acoustic or Schematic).

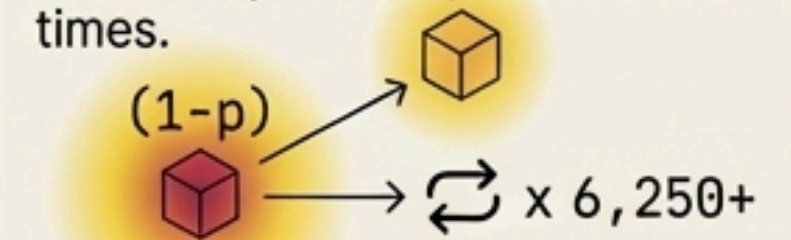
$$Y = X\beta + \epsilon$$

2. Regress

Apply Tikhonov-regularized Ridge Regression to predict the voxel pattern based strictly on the extracted musical features (Acoustic or Schematic).

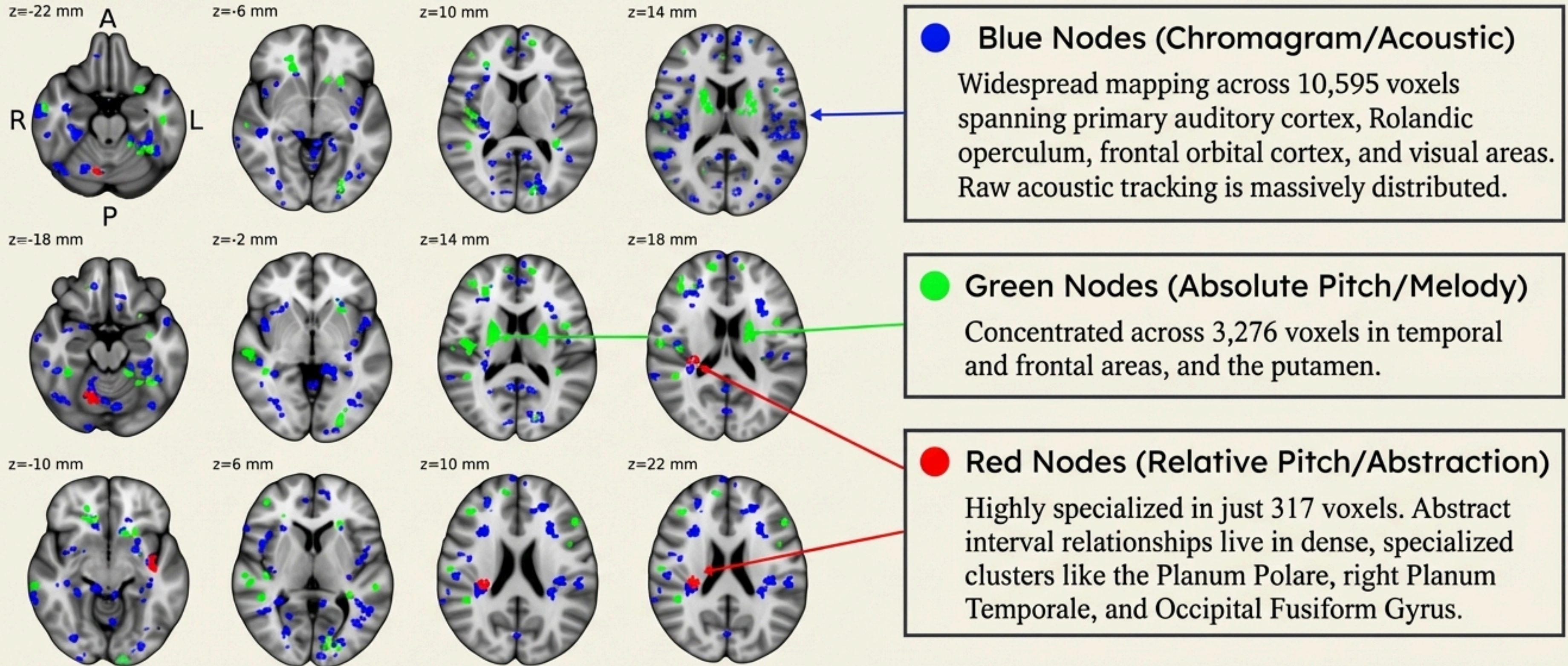

$$Y = X\beta + \epsilon$$

Assign a correlation-error probability score (1-p) to the center voxel, then randomly scatter the searchlight 3 voxels over to repeat the process 6,250+ times.



Context: This massive computational effort (15,000 CPU hours, 100,000 bootstrap samples) ensures we isolate exactly where specific musical features trigger predictive neural patterns.

The Cortical Topography of Music



Core Discovery: The brain processes complex polyphonic sound broadly, but distills abstract musical schemas (like relative pitch) into tiny, highly specialized neural networks.

The Geometric Architecture of Musical Experience

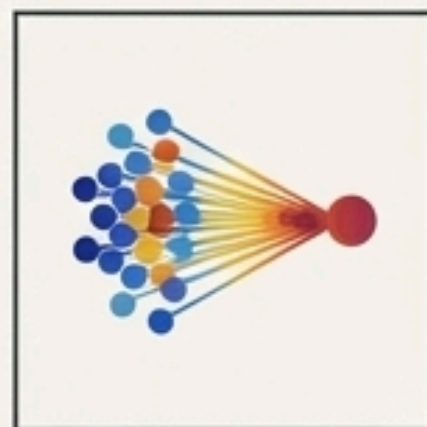
Distributed Acoustics

Raw sound profiles (timbre, audio energy) recruit massive, overlapping networks across temporal, frontal, parietal, and occipital lobes.



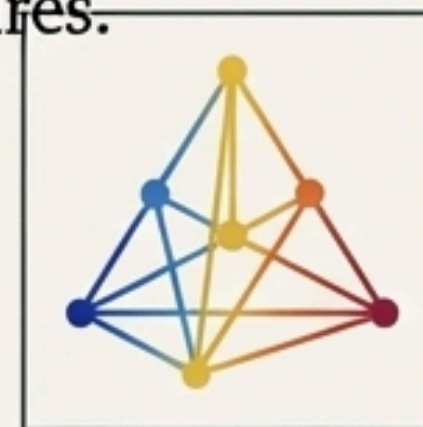
Specialized Abstraction

The structural grammar of music—relative pitch and melody schemas—is isolated in small, highly specialized computational nodes like the Planum Polare.



Behavioral Geometry

The multivoxel pattern spaces possess geometric properties. The mathematical distance between neural response vectors perfectly predicts how humans miscategorize music genres.



We do not just hear music; our brains construct high-dimensional geometric spaces to encode its grammar, categorize its style, and anticipate its resolution.

Thank You
